

EXERCISES

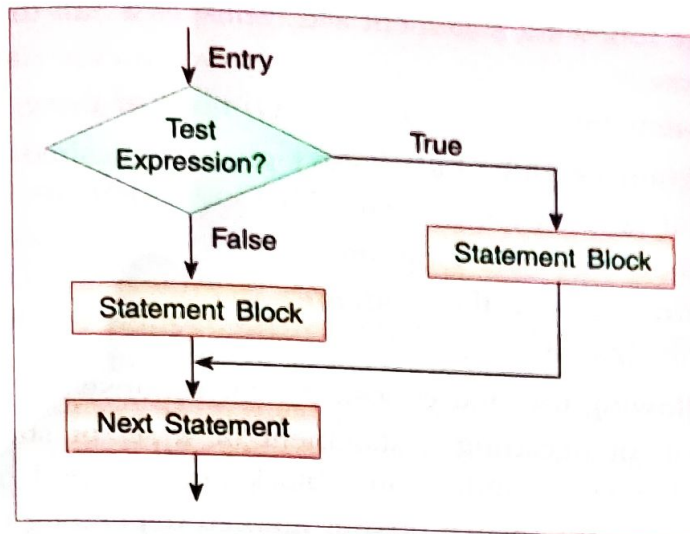


PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions

Tick (✓) the correct answer:

1.



Name the type of conditional statement in Java, depicted in the above flowchart.

- (a) if-else
 - (b) only if
 - (c) if-else-if
 - (d) nested if
2. In a switch case, when the switch value does not respond to any case then the execution transfers to:
- (a) a break statement
 - (b) a default case
 - (c) a loop
 - (d) none of these
3. Which of the following is a compound statement?
- (a) `p=Integer.parseInt(in.readLine( ));`
 - (b) `c=++a;`
 - (c) `if(a>b) a++; b-- ;`
 - (d) `a=4;`
4. Condition is essentially formed by using:
- (a) Arithmetic operators
 - (b) Relational operators
 - (c) Logical operators
 - (d) All of these
5. If $((a>b) \&\& (a>c))$, then which of the given statements is true?
- (a) b is the smallest number.
 - (b) b is the greatest number.
 - (c) a is the greatest number.
 - (d) All of these
6. `if(a>b)`
 `c=a;`
 `else`
 `c=b;`
It can be written as:
- (a) `c= (b>a)?a:b;`
 - (b) `c= (a!=b)?a:b;`
 - (c) `c= (a>b)?b:a;`
 - (d) `c= (a>b)?a:b;`
7. If a, b and c are the sides of a triangle then which of the following statement is true for: `if(a!=b && a!=c && b!=c)?`
- (a) Equilateral triangle
 - (b) Scalene triangle
 - (c) Isosceles triangle
 - (d) All of these

8. Two arithmetical expressions can be compared with the if statement, using
 - (a) Arithmetic operator
 - (b) Relational operator
 - (c) Logical operator
 - (d) None of these
9. Which of the following is a conditional statement?
 - (a) if
 - (b) goto
 - (c) for
 - (d) none of these
10. Which of the following statement accomplishes a 'fall through'?
 - (a) for statement
 - (b) switch case statement
 - (c) if-else statement
 - (d) none of these
11. A Java program executes but doesn't give the desired output. It is due to
 - (a) the logical error in the program
 - (b) the syntax error in the program
 - (c) the run time error in the program
 - (d) none of the above
12. Read the following text and choose the correct answer:
 It is a system of declaring a statement or a set of statements as a 'block'. The set of statements under same block are assumed to be a part of their parent statement and are executed or ignored depending on whether the given condition in the parent statement is 'True' or 'False'. In Java programming, it is maintained by using curly brackets.
 What is term defined for this feature in Java?
 - (a) Inclination
 - (b) Column
 - (c) Alignment
 - (d) Indentation

II. Assertion and Reasoning based questions:

Assertion (A): The function System.exit(0) is used to indicate the successful termination of the current program.

Reason (R): It exits current program by terminating the running Java Virtual Machine.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

III. Application based questions:

As homework assignment, the computer teacher has given a program snippet to check whether a non-zero integer input from the console is positive or negative. If the number is positive then it displays "Even Integer" otherwise, it displays "Odd Integer". In the snippet, she has left some places marked blank with ?1?, ?2?, ?3? and ?4? to be filled with appropriate value/statement. The incomplete snippet is as given below:

```
Scanner .....?1?..... = new Scanner(System.in);
System.out.println("Enter a non-zero integer:");
```



```

n = inp.nextInt( )
if(.....?2?.....)
    System.out.println("Negative Integer");
else if(n % 2 == 0)
    p = 1;
else
    p = .....?3?.....;
switch(p)
{
    case 1: System.out.println(.....?4?.....);
            break;
    case 2: System.out.println("Odd Integer");
            break;
}

```

Based on the above snippet, answer the following questions:

- Name the object to be filled in place of ?1?
- Write the condition to be filled in place of ?2?
- What value will be filled in ?3? that should be assigned to p?
- What message will you write in place of ?4?

IV. Predict the output of the given snippets, when executed:

1. int a=1,b=1,m=10,n=5;

```
if((a==1)&&(b==0))
```

```
{
```

```
    System.out.println((m+n));
```

```
    System.out.println((m-n));
```

```
}
```

```
if((a==1)&&(b==1))
```

```
{
```

```
    System.out.println((m*n));
```

```
    System.out.println((m%n));
```

```
}
```

2. int x=1,y=1;

```
if(n>0)
```

```
{
```

```
    x=x+1;
```

```
    y=y+1;
```

```
}
```

What will be the values of x and y, if n assumes a value (i) 1 (ii) 0?

[ICSE-2009]

3. int b=3,k=0,r;

```
double a=15.15,c=0;
```

```
if(k==1)
```

```
{
```

```
    r=(int)a/b;
```

```
    System.out.println(r);
```

```
}
```

```
else
```

```

    {
        c=(double)a/b;
        System.out.println(c);
    }
4. switch(opn)
{
    case 'a':
        System.out.println("Platform Independent");
        break;
    case 'b':
        System.out.println("Object Oriented");
    case 'c':
        System.out.println("Robust and Secure");
        break;
    default:
        System.out.println("Wrong Input");
}

```

When (i) opn = 'b' (ii) opn = 'x' (iii) opn = 'a'

5. Predict the output when x='g':

```

switch(x)
{
    case 'M':
        System.out.print("Microsoft Teams");
        break;
    case 'G':
        System.out.print("Google Meet");
    default:
        System.out.print("Any software");
        break;
    case 'W':
        System.out.print("Web Ex");
        break;
}

```

V. Correct the errors in the given programs:

1. class public

```

{
    public static void main(String args{ })
    {
        int a=45,b=70,c=65.45;
        sum=a+b;
        diff=c-b;
        System.out.println(sum,diff);
    }
}

```

2. class Square

```
{  
    public static void main(String args[ ])  
    {  
        int n=289,r;  
        r=sqrt(n);  
        if(n==r)  
            System.out.println("Perfect Square");  
        else  
            System.out.println("Not a Perfect Square");  
    }  
}
```

3. class Simplify

```
{  
    public static void main(String args[ ])  
    {  
        int a,b,c,d;  
        a=10,b=5,c=1,d=2;  
        c=a2+b2;  
        d=(a+b)2;  
        p=c/d;  
        System.out.println(c + " " + " "+d+ " "+p);  
    }  
}
```

4. class Sample

```
{  
    public static void main(String args[ ])  
    {  
        int n,p;  
        float k,r;  
        n=25;p=12;  
        if(n=25)  
        {  
            k=pow(p,2)  
            System.out.println("The value of"+p+ " = "+k);  
        }  
        else  
        {  
            r=Math.square root(n);  
            System.out.println("The value of"+n+ " = "+r);  
        }  
    }  
}
```


VI. Unsolved Java Programs:

1. Write a program to input three angles of a triangle and check whether construction is possible or not. If possible then check and display whether it is an acute-angled triangle, right-angled triangle or an obtuse-angled triangle. Otherwise, display the message "A Triangle is not possible".
Sample Input: Enter three angles: 40, 50, 90
Sample Output: Right-Angled Triangle
2. Write a program to input the cost price and the selling price of an article. If the selling price is more than the cost price then calculate and display the actual profit and profit per cent otherwise, calculate and display the actual loss and loss per cent. If the cost price and the selling price are equal, the program displays the message "Neither profit nor loss".
3. Write a program to input three numbers and check whether they are equal or not. If they are unequal numbers then display the greatest among them otherwise, display the message "All the numbers are equal".
Sample Input: 34, 87, 61
Sample Output: Greatest number: 87
Sample Input: 81, 81, 81
Sample Output: All the numbers are equal.
4. Write a program to accept a number and check whether the number is divisible by 3 as well as 5. Otherwise, decide:
(a) Is the number divisible by 3 and not by 5?
(b) Is the number divisible by 5 and not by 3?
(c) Is the number neither divisible by 3 nor by 5?
The program displays the message accordingly.
5. Write a program to input the year. The program checks and displays whether it is (a) a Leap year (b) a Century Leap year (c) a Century year but not a Leap year
Sample Input: 2000
Sample Output: It is a Century Leap Year.
6. Write a program to input two unequal positive numbers and check whether they are perfect square numbers or not. If the user enters any negative number then the program displays the message "Square root of a negative number cannot be determined".
Sample Input: 81, 100
Sample Output: They are perfect square numbers.
Sample Input: 225, 99
Sample Output: 225 is a perfect square number.
99 is not a perfect square number.
7. Without using if-else statement and ternary operators, accept three unequal numbers and display the second smallest number.
[Hint: Use Math.max() and Math.min() functions]
Sample Input: 34, 82, 61
Sample Output: 61

8. Write a program to input three unequal numbers. Display the greatest and the smallest number.

Sample Input: 28, 98, 56

Sample Output: Greatest Number: 98
Smallest Number: 28

9. A Prepaid taxi charges from the passenger as per the tariff given below:

Distance	Rate
Up to 5 km	₹ 100
For the next 10 km	₹ 10/km
For the next 10 km	₹ 8/km
More than 25 km	₹ 5/km

Write a program to input the distance covered and calculate the amount to be paid by the passenger. The program displays the printed bill with the details given below:

Taxi No. : _____

Distance covered : _____

Amount : _____

10. A cloth showroom has announced festival discounts and the gifts on the purchase of items, based on the total cost as given below:

Total Cost	Discount	Gift
Up to ₹ 2,000	5%	Calculator
₹ 2,001 to ₹ 5000	10%	School Bag
₹ 5,001 to ₹ 10,000	15%	Wall Clock
Above ₹ 10,000	20%	Wrist Watch

Write a program to input the total cost. Compute and display the amount to be paid by the customer along with the gift.

11. Given below is a hypothetical table showing the rate of income tax for an Indian citizen, who is up to the age of 60 years.

Taxable Income (TI) in ₹	Income Tax in ₹
Up to ₹ 2,50,000	Nil
More than ₹ 2,50,000 and less than or equal to ₹ 5,00,000	$(TI - 1,60,000) * 10\%$
More than ₹ 5,00,000 and less than or equal to ₹ 10,00,000	$(TI - 5,00,000) * 20\% + 34,000$
More than ₹ 10,00,000	$(TI - 10,00,000) * 30\% + 94,000$

Write a program to input the name, age and taxable income of a person. If the age is more than 60 years then display the message "Wrong Category". If the age is less than or equal to 60 years then compute and display the income tax payable along with the name of the tax payer, as per the table given above.

[ICSE-2012]

12. A company announces revised Dearness Allowance (DA) and Special Allowance (SA) for their employees as per the tariff given below:

Basic	Dearness Allowance (DA)	Special Allowance (SA)
Up to ₹ 10,000	10%	5%
₹ 10,001 – ₹ 20,000	12%	8%
₹ 20,001 – ₹ 30,000	15%	10%
₹ 30,001 and above	20%	12%

Write a program to accept the name and Basic Salary (BS) of an employee. Calculate and display the gross salary.

Gross Salary = Basic + Dearness Allowance + Special Allowance

Display the information in the given format:

Name	Basic	DA	Spl. Allowance	Gross Salary
xxx	xxx	xxx	xxx	xxx

13. Using a switch case statement, write a menu driven program to convert a given temperature from Fahrenheit to Celsius and vice-versa. For an incorrect choice, an appropriate message should be displayed.

[Hint: $c = \frac{5}{9} * (f - 32)$ and $f = 1.8 * c + 32$]

[ICSE-2007]

14. The volume of solids viz. cuboid, cylinder and cone can be calculated by the formulae:

1. Volume of a cuboid ($v = l * b * h$)

2. Volume of a cylinder ($v = \pi * r^2 * h$)

3. Volume of a cone ($v = \frac{1}{3} * \pi * r^2 * h$)

$[\pi = \frac{22}{7}]$

Using a switch case statement, write a program to find and display the volume of different solids by taking suitable variables and data types as input.

15. A Mega Store has different floors which display varieties of dresses as mentioned below:

1. Ground floor : Kids Wear
2. First floor : Ladies Wear
3. Second floor : Designer Sarees
4. Third Floor : Men's Wear

The user should enter the floor number to get the information regarding different items of the Mega Store. After shopping, the shopkeeper inputs the item purchased and cost of the item and prints a bill to be paid by the customer in the given format:

Name of the Store : City Mart

Total Amount : _____

Visit Again!!!

Write a program to perform the above task as per the user's choice.

16. The equivalent resistance of series and parallel connections of two resistances is given by the formula:

(a) $R_1 = r_1 + r_2$ (Series)

(b) $R_2 = \frac{r_1 * r_2}{r_1 + r_2}$ (Parallel)

Using a switch case statement, write a program to enter the values of r_1 and r_2 . Calculate and display the equivalent resistance accordingly.

17. The Simple Interest (SI) and Compound Interest (CI) of a sum (P) for a given time (T) and rate (R) can be calculated as:

$$(a) SI = \frac{P \cdot r \cdot t}{100} \quad (b) CI = P \left(\left(1 + \frac{R}{100} \right)^T - 1 \right)$$

Write a program to input sum, rate, time and type of Interest ('S' for Simple Interest and 'C' for Compound Interest). Calculate and display the sum and the interest earned. Use switch case statement.

18. 'Kumar Electronics' has announced the following seasonal discounts on purchase of certain items.

Purchase Amount	Discount on Laptop	Discount on Desktop PC
Up to ₹ 25000	0.0%	5.0%
₹ 25,001 to ₹ 50,000	5.0%	7.5%
₹ 50,001 to ₹ 1,00,000	7.5%	10.0%
More than ₹ 1,00,000	10.0%	15.0%

Write a program to input name, amount of purchase and the type of purchase ('L' for Laptop and 'D' for Desktop) by a customer. Compute and display the net amount to be paid by a customer along with his/her name.

(Net amount = Amount of purchase – Discount)

[ICSE-2009]

PART B – SUBJECTIVE QUESTIONS

I. Answer the following questions:

- Name the different ways to manage the flow of control in a program.
- Mention one statement each to achieve:
 - Conditional flow of control
 - Multiple branching of control
- Explain the following statements with their constructs:
 - nested if
 - if – else
 - if – else – if
- Differentiate between the if and switch statement. [ICSE-2005]
- What is the purpose of the switch statement in a program?
- Explain with the help of an example, the purpose of default in a switch statement. [ICSE-2005]
- Is it necessary to use the 'break' statement in a switch case statement? Explain.
- Explain 'Fall through' with reference to a switch case statement.
- What is a compound statement? Give an example. [ICSE-2005]
- Explain, with an example, the if-else-if construct. [ICSE-2007]
- What is the significance of System.exit(0)?
- Give two differences between the switch statement and the if-else statement. [ICSE-2014]